

edx Phot1x report template (2026/07)

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the date of receipt and acceptance should be inserted later

Abstract The report is about an exercise proposed by the edX translation UBCx Phot1xSilicon Photonics Design, Fabrication and Data Analysis. This report is based on Ansys Interconnect for circuit simulation, Luceda Ipikiss software for layout/schematic and circuit simulation and klayout for layout visualization and DRC.

1 Introduction

Silicon photonics enables the miniaturization of free-space optical systems by integrating optical components onto a silicon chip. This approach can be effectively applied to optical sensing technologies, which have become essential in modern engineering due to their ability to provide precise, distributed, and real-time measurements of physical parameters. Among these technologies, interferometric methods are particularly prominent. For instance, the Michelson interferometer operates by splitting a light beam into two separate paths, reflecting them back, and recombining them to generate interference patterns. These patterns contain detailed information about variations in optical path length caused by changes in distance, refractive index, or external perturbations such as strain and temperature. This same interferometric principle underpins optical frequency domain reflectometry, where the analysis of interference as a function of optical frequency enables spatially resolved sensing along an optical fiber.

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2 Theory

2.1 Modeling

Beginning with the physics behind a y-branch splitter, let's consider an input intensity I_i and input electric field E_i . At the light is splitted at the output of this building-block, thus the intensity became $I_1 = I_2 = \frac{I_i}{2}$ and the electric field $E_1 = E_2 = \frac{E_i}{\sqrt{2}}$. For the combiner, we have $E_1 = \frac{E_{i1}}{\sqrt{2}}$ and $E_{i2} = \frac{E_i}{\sqrt{2}}$ which results $E_o = \frac{E_{i1} + E_{i2}}{\sqrt{2}}$

Considering a plane wave $E = E_0 e^{i(\omega t - \beta z)}$ with the propagation constant $\beta = \frac{2\pi n}{\lambda}$

$$E_{i1} = E_1 e^{-i\beta_1 L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1} \quad (1)$$

$$E_{i2} = E_2 e^{-i\beta_2 L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2} \quad (2)$$

Hence the output of the combiner:

$$E_o = \frac{E_{i1} + E_{i2}}{\sqrt{2}} = \frac{E_i}{2} (e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}) \quad (3)$$

And the intensity is:

$$I_o = \frac{I_i}{4} |e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}| \quad (4)$$

$$I_o = \frac{I_i}{4} (1 + \cos(\beta \Delta L)) \quad (5)$$

Using the siepic pkg from Luceda ipikiss, we ensure that the strip waveguide used has the following dimensions of width 500nm and heigh 220nm (cf figure 1).

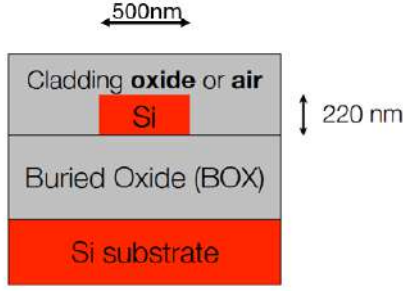
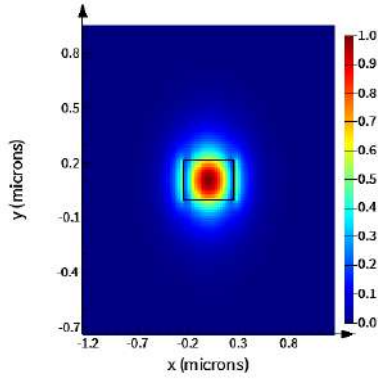


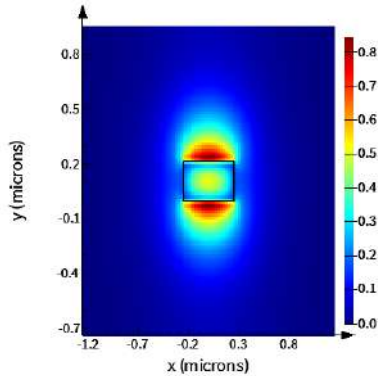
Fig. 1

Cross section model of the Si waveguide (500x220nm)

Then we simulate the fundamental modes allowed to propagate in this waveguide, presented in the following figure 2a and 2b respectively the quasi-TE0 and quasi-TM0 fundamental modes.



(a)



(b)

Fig. 2: (a) quasi-TE mode (b) quasi-TM mode in Si waveguide

2.1.1 Effective index and group index versus wavelength

A wavelength sweep in Lumerical MODE was performed on the silicon (Si) waveguide model defined previously (see Figure 1). This sweep was conducted from 1.5 μm to 1.6 μm to obtain the material dispersion curve. The resulting data points were exported to a CSV file and subsequently extracted to apply a polynomial fit, allowing the retrieval of the Sellmeier model coefficients (Equation 2.1.1).

$$n_{eff}(\lambda_0) = 2.44 - 1.13 \cdot (\lambda - \lambda_0) - 0.06 \cdot (\lambda - \lambda_0)^2 \quad (6)$$

with $\lambda_0 = 1.55 \mu\text{m}$

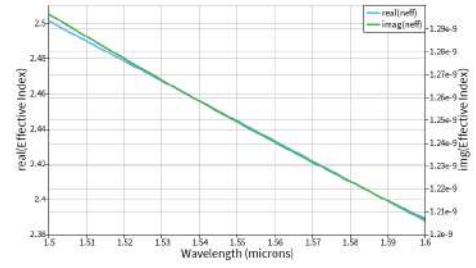
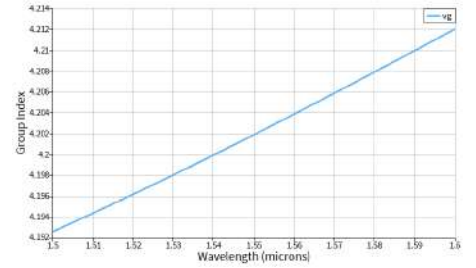
(a) Effective index n_{eff} of quasi-TE mode.(b) Group index n_g of quasi-TE mode.

Fig. 3: Quasi-TE mode characteristics as a function of wavelength: (a) effective index, and (b) group index where $n_g = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda}$.

2.1.2 Insertion loss of Grating Couplers

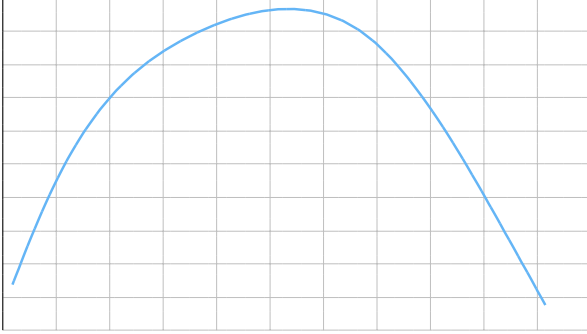


Fig. 4

Insertion loss of the grating coupler

The insertion loss of Grating Couplers 4 is an important loss to take in account because this the transfer function which be the responsible of the final shape of the signal, as shown in 5, where the MZI signal in blue become the one in green when the insertion loss of the input and output grating are added on this signal (green).

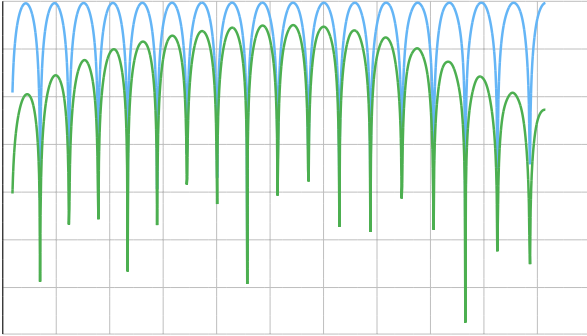


Fig. 5

MZI spectrum with (in green) or without grating couplers (in blue)

2.1.3 Mach-Zehnder Interferometer (MZI)

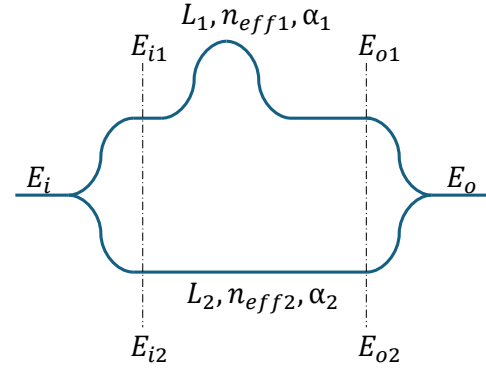


Fig. 6: Schema of a Mach-Zehnder interferometer

Equation for an unbalanced interferometer with identical waveguides:

$$I_0 = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (7)$$

In an MZI, light enters one port, splits, travels through the two arms once, and recombines at the output.

Physical length difference: $\Delta L = L_1 - L_2$

Total path difference: Since light only goes through the arms one time, $\Delta_{total} = \Delta L = L_1 - L_2$.

FSR Formula:

$$FSR_{MZI}(\lambda) = \frac{\lambda^2}{n_g \Delta L} \quad (8)$$

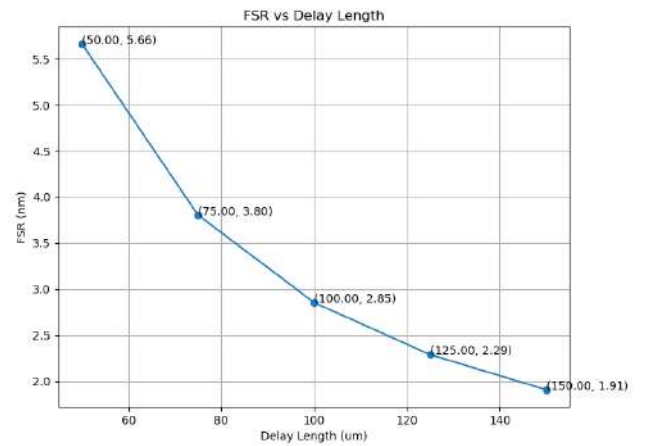


Fig. 7

FSR vs delay length for bend radius of 5um for MZI.

2.1.4 Michelson Interferometer

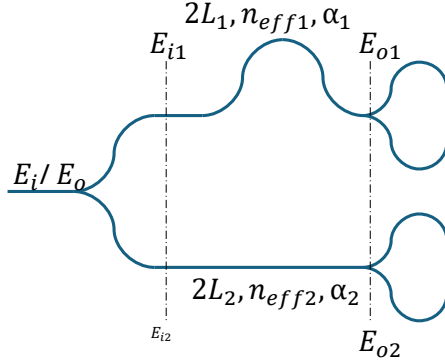


Fig. 8: Schema of a michelson interferometer

In a Michelson interferometer, light splits, travels down the arms, hits mirrors, and returns back through the same arms to recombine.

Physical length difference: $\Delta L = L_1 - L_2$.

Total path difference: Because light travels the arms twice (there and back), the effective difference is doubled: $\Delta_{total} = 2\Delta L = 2(L_1 - L_2)$.

FSR Formula:

$$FSR_{Michelson}(\lambda) = \frac{\lambda^2}{n_g 2\Delta L} \quad (9)$$

2.2 Simulations

2.2.1 Michelson Interferometer

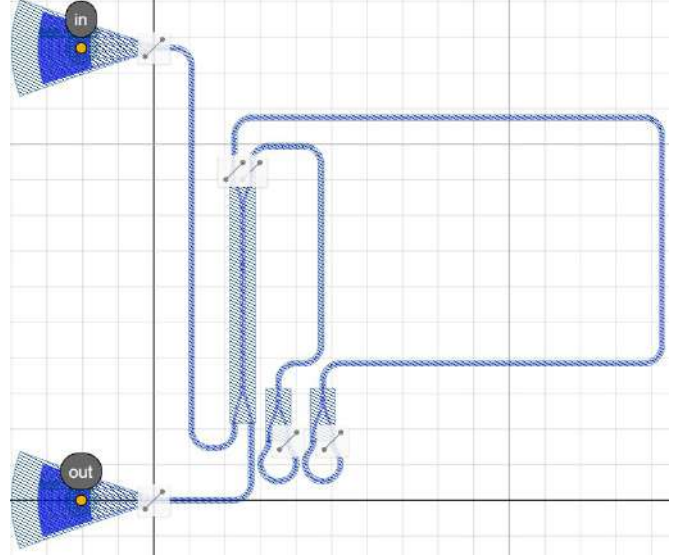


Fig. 9

Michelson Interferometer in Luceda software schematic with 125 μm length delay.

2.2.2 Notes about the different layouts

- layout **A**: Michelson interferometer with different length delay and a compact directional coupler. All component working at TE polarization for centered wavelength 1550 nm.

A WBG temperature sensor has been added to the layout for fun.

- layout **B**: Michelson interferometer with different length delay and a adiabatic directional coupler. All component working at TE polarization for centered wavelength 1550 nm.

- layout **C**: Attemp to make a Polarization-Splitter Rotator, inspired by the layout of this PolarizationSplitterRotator. The issues are that siepic pdk doesn't propose SiO2 cladding, necessary for this component to work and the other issue are pure layout mismatch between the width of the PSR's waveguides and the siepic waveguide section.

Tests structures of other devices has been added.

3 Fabrication

Three chips were fabricated in this course, corresponding to the 3 different layout mentioned previously (A,B,C).

3.1 Layouts

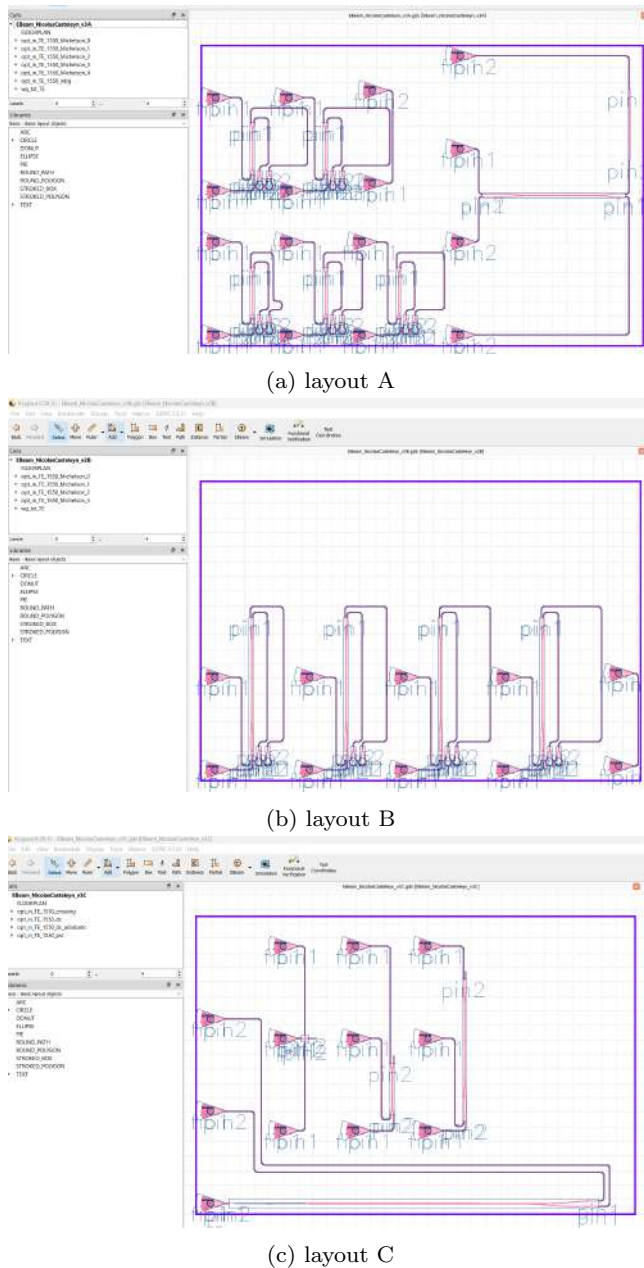


Fig. 10

3.2 Washington Nanofabrication Facility (WNF) silicon photonics process:

The devices were fabricated using 100 keV Electron Beam Lithography [2]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 μm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 $^{\circ}\text{C}$ for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 μm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 $\mu\text{C}/\text{cm}^2$ was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 20 $^{\circ}\text{C}$, resulting in a bias voltage of 185 V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil.

3.3 Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure

of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A $2.2\ \mu\text{m}$ oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C . Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

3.4 Corner Analysis

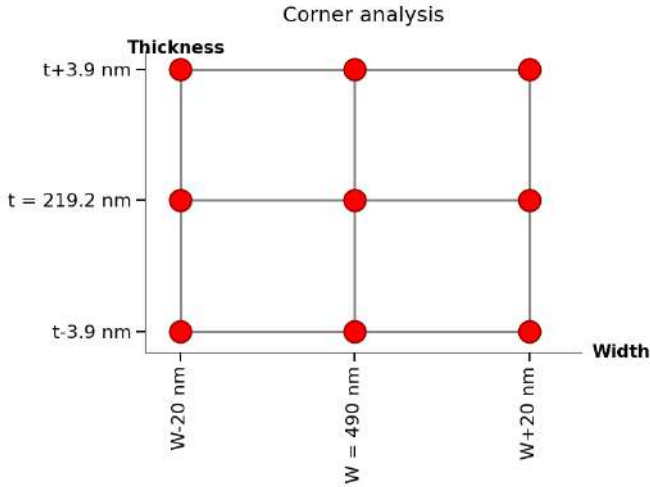
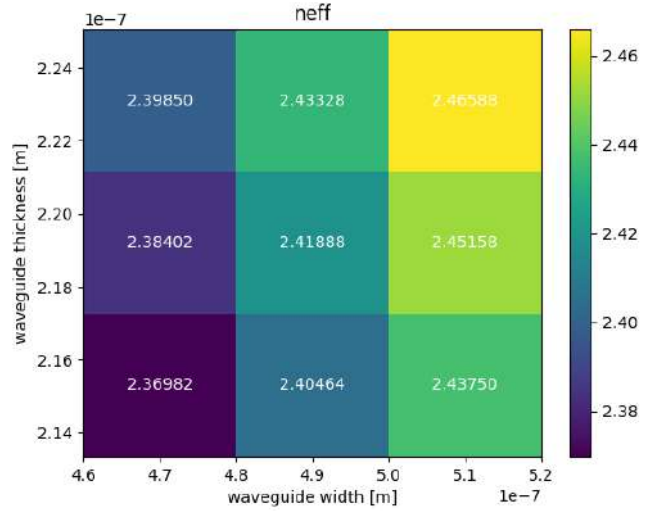
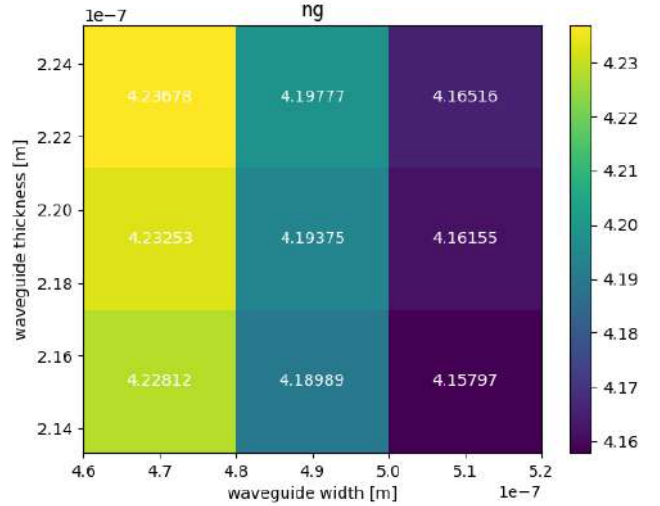


Abb. 11

Grid for corner analysis



(a) n_{eff} corner analysis



(b) n_g corner analysis

Abb. 12: Corner analysis of the waveguide parameters for (a) effective index and (b) group index.

4 Experimental Data

To characterize the devices, a custom-built automated test setup [[3]] with automated control software written in Python was used (<http://siepic.ubc.ca/probestation>). An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [[4]]. A 90° rotation was used to inject light into the TM grating couplers [4].

A polarization maintaining fibre array was used to couple light in/out of the chip [www.plcconnections.com]

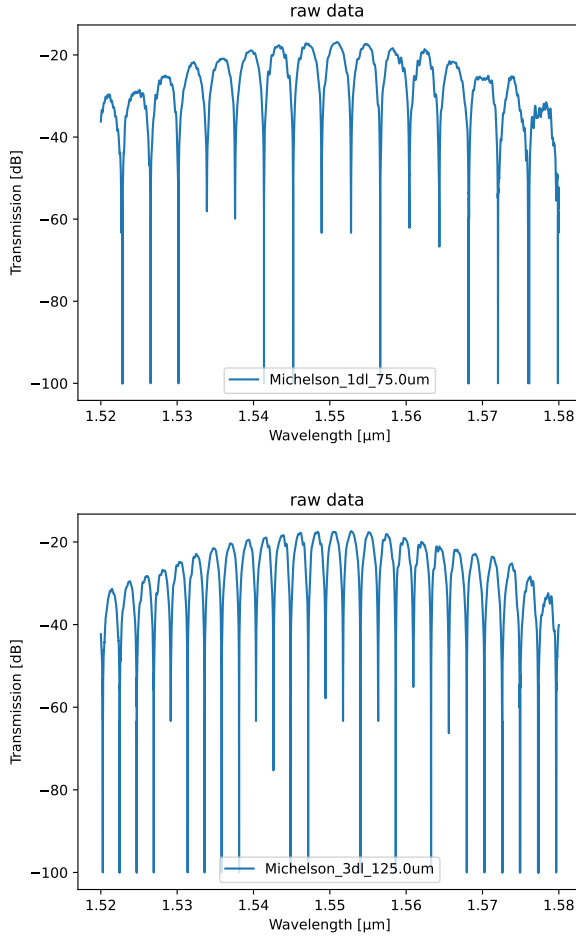


Abb. 13: Measured data on the range 1.5-1.6 μm

5 Analysis

5.1 Measure the group index

From the raw data [figure 13] is applied a baseline correction in order to avoid the transfer matrix of the grating coupler and get all the lobes of the signal aligned on 0 dBm. Then, a 1d interpolation is applied on the corrected amplitude and the wavelength vector is redefined to have a regular step. The resulting signal is then smoothed to perform a peak detection to estimate the FSR of the signal.

From this first evaluation of FSR is extracted a estimated group index and dispersion. Those estimated values will be used to use our mode of michelson transfert

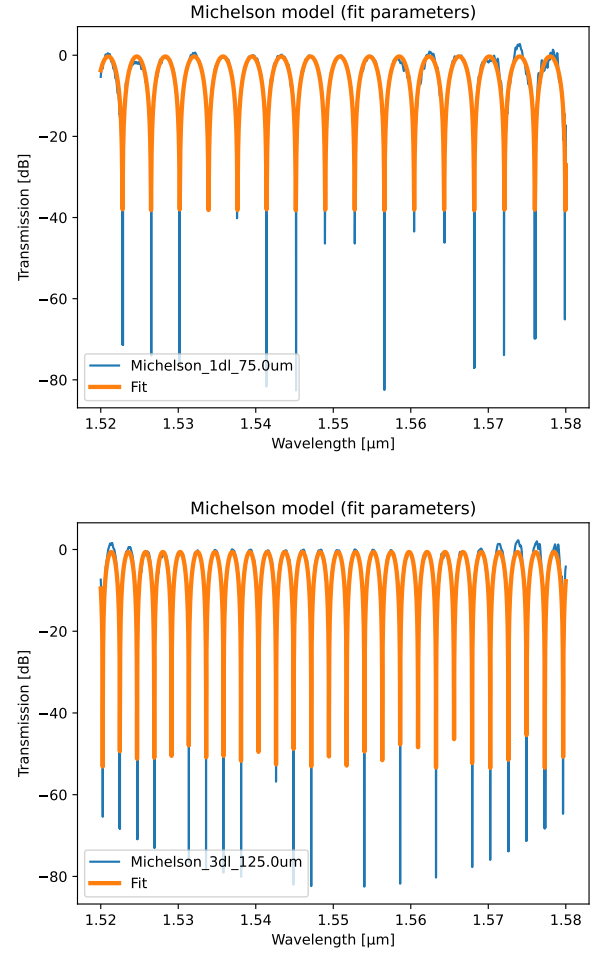


Abb. 14: Data analysis to extract the waveguide group index, etc.

matrix in order to fit with the measured data. Once the polifit alorithm fits the transfert function with the experimental data, the measured group index and dispersion of the interferometer can be extracted as shown in table 1.

Tabelle 1: Experimental parameters for the Michelson interferometer configurations.

Michelson	Length Delay (μm)	Group Index (n_g)
1	75	4.2107
2	100	4.2030
3	125	4.2044

5.2 Comparison of experimental results with simulations with Luceda

Since Luceda has been used to make the layouts, the simulation tool has been used too. A find peak algorithm is performed on the simulated signal of each interferometer understood as theoretical values. Thus, the experimental and theoretic values are compared in the following plot 15

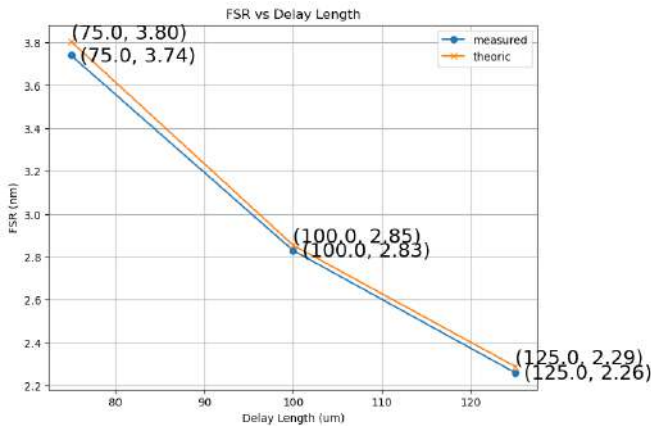


Abb. 15

Measured FSR delay vs theoretic

6 Conclusion

To conclude, all the measured FSR (table 1) are in the range of the corner analysis previously made (figure 12), between the values $4.15797 < n_g < 4.23678$. Which is a coherent analysis with the simple FSR estimation extracted from find peak algorithm in plot 15.

To finish, I had a lot of fun performing the edX training, by discovering tools, making mistakes etc...

7 Acknowledgements

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